

BRAC University
CSE 350 (Summer 2025)
Section 17
Quiz-1

Name :

ID :

Problem 1:

(10 Marks)

Design a 3-bit Flash ADC for a signal with maximum & minimum value of **8V** & **-3V** respectively. Use **Mide-tread** uniform quantization. Answer the following:.....

- (a) Draw the circuit diagram with minimum hardware configuration and compute the Quantization table.
- (b) If for specific input **V_{in}** the encoded output is **101**, then determine the possible range that **V_{in}** lies in.

Problem 2:

For this **6 bit** Dual Slope ADC, $R=25\text{k}\Omega$,
 $C=2\mu\text{F}$.

(a) Determine **Vref** and If the sampled and held input voltage to the ADC **V_a**, then determine the **total time** it takes to encode this **V_a**. (Given , **T_{clock}** = **2ns**).

(10 marks)

